

Chromatography: A substance being analyzed (the analyte), the mobile phase, the stationary phase is a solid material which the mobile phase travels through. If a part of the mixture interacts strongly with the stationary phase, it will separate out of the mixture more quickly. If a part of the mixture interacts weakly with the stationary phase, it will travel further from the point of origin. As a result, the mixture is separated based on a particular property.

Retention factor: $R_f = \text{Dp/Dv}$ = distance from pigment/distance from solvent (Measure the solvent starting from where the pigment begins, not the bottom of the paper/plate. Also, it's always between 0 and 1, and a pigment that travels the same as solvent would be $R_f = 1$). A value close to 1 means that the substance has a high affinity for the mobile phase, and a R_f value close to 0 means that the substance has a high affinity for the stationary phase)

Paper chromatography: The stationary phase is the cellulose in the paper, which is very polar. The mobile phase is usually a non-polar solvent like rubbing alcohol. Non-polar substances will interact strongly with the mobile phase and travel far up the paper, while polar substances will react with the mobile phase and stay close to the point of origin.

Performing Paper Chromatography: Paper chromatography separates individual pigments/components. You need to suspend the filter paper with the rod above the solvent in the beaker. You should use about 50 mL of solvent.

Take the paper strip and draw a line in pencil (not pen) about 1 cm from the bottom of the strip. Place a dot in the center of pencil line. Next, fill the beaker with the desired solvent so that the solvent stops just before the pencil line on the paper. It is important to not submerge the dot in the solvent, as the chromatography will not work if this happens. Attach the top of the strip of filter paper to a dowel or rod using a binder clip and place the bottom of the filter paper into the beaker. The dowel or rod will suspend the paper in the solvent so it travels up the filter paper and separates the pigments in the ink. Once the pigments have stopped moving (or reached the top of the paper), remove the paper from the beaker and mark where the solvent stopped.

Interpreting paper results: Measure the maximum distance traveled by each pigment starting at the line. More soluble in solvent or less mass = travels farther

Thin-layer chromatography: TLC utilizes a thin plate made of a nonreactive substance like glass covered in an adsorbent material like silica (SiO_2) or alumina (Al_2O_3). The silica serves as the stationary phase, adhering to the material being separated by the solvent.

Performing TLC: A TLC plate has two sides: smooth/shiny side and matte side with silica deposit. Draw

Performing thin layer chromatography is similar to paper chromatography. A TLC plate has two sides: a smooth, shiny side and a matte side with the silica deposit. The silica is very porous and polar. Draw a line in pencil on the matte side of the plate around 1 cm from the bottom. This may disturb the silica layer, but that won't affect the results of the chromatography. Be careful when handling the plate so as to not snap it. Then, place a dot of ink on the pencil line. If the substance being analyzed is a liquid, it may be necessary to use a thin tube called a capillary or a toothpick to place the analyte on the plate. Touch the capillary/dropper into the substance, then touch it to the plate briefly. Then, treat the plate like it were a piece of paper in paper chromatography. Pour solvent into the beaker and place the TLC plate in it, leaning the plate up against the edge of the beaker. Since the plate is rigid, a dowel or rod isn't necessary to suspend it. Note that the solvent must be less polar than the silica gel.

Interpreting TLC results: Measure the maximum distance traveled starting at the line. Less polar = travels farther

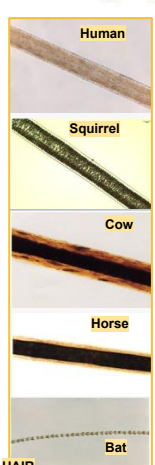
US burn test density guide to identify polymers

Plastic	Abbreviation	Density	Monomer Unit Structure	Other Key Features	Commonly Used to Make	Angle
Polystyrene	PS	1.05 g/cm ³	$-\text{CH}_2-\text{CH}(\text{C}_6\text{H}_5)-$	- yellow flame - burns quickly - plastic drips - flaming gas odor (hydrocarbons) - dense black smoke w/ soot (flying particles)	Polymerizes by addition, styrofoam, tableware, coffee cups, toys, lighting signs, insulation	
Polypropylene	PP	0.90 g/cm ³	$-\text{CH}_2-\text{CH}(\text{CH}_3)-$	- blue, yellow tipped flame - burns - plastic - flaming gas odor - melts in water	Polymerizes with addition food containers, medicine containers, Automobile Batteries, carpet, rope, plastic wrap, lab equipment	
Polyvinyl Chloride	PVC	1.36 g/cm ³	$-\text{CH}_2-\text{CH}(\text{Cl})-$	- yellow flame w/ green soot - self-extinguishes - smells like HCl gas	Burns green, polymerizes by addition Food packaging, shampoo containers, Construction (shem PVC pipes - you see them often), Tiles, credit cards	
Low Density Polyethylene	LDPE	0.92 g/cm ³	$-\text{CH}_2-\text{CH}_2-$	- white, yellow flame - melts - smells like candle wax	Polymerizes by addition, Monomer units branch out more Bags, plastic wrap, etc	
High Density Polyethylene	HDPE	0.95 g/cm ³	$-\text{CH}_2-\text{CH}_2-$	- white, yellow flame - melts - smells like candle wax	Polymerizes by addition, Monomers more linear, flower pots signs, trash cans, toys.	
Polycarbonate	PC	1.20 g/cm ³	$-\text{O}-\text{C}_6\text{H}_4-\text{O}-\text{C}(\text{CH}_3)_2-\text{C}_6\text{H}_4-\text{O}-$	- orange flame - hard, brittle - black smoke w/ soot in air	Polymerizes by Condensation, shatters waterproof glass, eyeglass lenses	
Polyethylene Terephthalate	PETE	1.37 g/cm ³	$-\text{O}-\text{C}_6\text{H}_4-\text{O}-\text{C}_6\text{H}_4-\text{C}(\text{O})-\text{O}-$	- light blue flame, pop is white - melts - con	Polymerizes by Condensation, shrivels heat soft drink bottles, carpet, fiberfill, food, scouring pads, fabric, mylar	
Polymethyl Methacrylate	PMMA	1.16 g/cm ³	$-\text{CH}_2-\text{C}(\text{CH}_3)(\text{COOCH}_3)-$	- light blue flame, pop is white - melts - con	Polymerizes by addition Reacts with acetone plexiglass, glass substitute	

Name of Fiber	Type of Fiber	Fact About Fiber Type	Burn Test Results	Microscopic View
Wool	Animal	Most commonly used animal fiber	shrivels, leaves brown-black residue, smells like burning hair	hair cylinder with scales
Silk	Animal	Smoother than wool	mainly consists of 2 proteins fibroin and sericin shrivels, leaves black residue, smells like burning hair	thin, long and smooth cylinder
Cotton	Vegetable	Most widely used plant fiber, fairly short fibers, $\text{C}_6\text{H}_{10}\text{O}_5$ fibers generally longer and smoother than cotton	burns with a steady flame, smells like burning paper, able to blow flame from thread like a match, leaves a charred whitish ash	irregular twisted ribbon
Linon	Vegetable	fibers generally longer and smoother than cotton	burns at a constant rate, does not produce smoke, smells like burning grass, produces sparks	smooth, bamboo like structure
Polyester	Synthetic	fibers can be any length	melts, only ignites when in the flame, drops when it burns and bonds quickly to any surface it drips on, produces sweet odor and hard, colored (same as fiber) ash	completely smooth cylinder plastic
Nylon	Synthetic	long fibers, $\text{C}_6\text{H}_{10}\text{O}_5$ diamine can stretch to eight times its original length	curls, melts, produces black residue, smells like burning plastic (some sources say it smells like celery?), ignites only when brought into flame	fine, round, smooth, translucent
Spandex	Synthetic	can stretch to eight times its original length	melts quickly	flattened, rod-like fibers, clusters

HOW TO READ YOUR RESULTS

BLOOD TYPE	ANTI-A	ANTI-B	ANTI-D	CONTROL
O-POSITIVE	+	+	+	+
O-NEGATIVE	+	+	-	+
A-POSITIVE	+	-	+	+
A-NEGATIVE	+	-	-	+
B-POSITIVE	-	+	+	+
B-NEGATIVE	-	+	-	+
AB-POSITIVE	+	+	+	+
AB-NEGATIVE	+	+	-	+
INVALID	+	+	+	-



HAIR

DNA fingerprints are created through a process known as gel electrophoresis. DNA molecules are placed into a box containing a buffer solution and gel (typically agarose), as well as positive and negative electrodes. DNA is placed into wells in the gel, and the current is turned on. Because the DNA is negatively charged, it will migrate through the gel towards the positive electrode. **Shorter DNA fragments travel further away** from the wells through a process known as sieving. The agarose gel is a matrix, and it is much easier for the short fragments to move through the pores in the gel. Long DNA fragments will get caught and tangled, and stop moving as a result. This separates the fragments, producing the complete fingerprint. To determine if an individual's DNA matches the DNA at the crime scene, simply match the bands on the fingerprints. If the bands don't match, the DNA isn't the same and the suspect was not the one whose DNA was found at the crime scene. However, the DNA used in gel electrophoresis has to come from somewhere.

Polymerase Chain Reaction (PCR) is a method of synthetic DNA replication used to amplify fragments of DNA for use in fingerprinting. It has three steps: denaturing, annealing, and synthesis/elongation. In denaturing, the DNA is heated to approximately 95 °C in order to break apart the two strands so that new DNA can be created. In the annealing stage, the DNA is cooled to anywhere between 50 and 56 °C so that a primer can be attached to the DNA strands. This primer is essential for the replication of DNA, which occurs in the final stage. During synthesis/elongation, new DNA is created using the primer as a starting point. An enzyme known as Taq polymerase creates a new DNA strand, using the starting strand as a template.

Bullet Striations: Microscopic ridges on the surface of a bullet. They are created when the bullet exits the barrel of a gun. Each gun barrel has unique grooves known as **rifling**. The purpose of this rifling is to give the bullet a spin as it travels, which improves the spin on the weapon. Since the barrel is harder than the bullet, these ridges will imprint on the surface of the bullet when the gun is fired. Since the rifling of every gun barrel is unique, it is possible to match the grooves between a bullet and a gun to see which gun the bullet was fired from. However, not all guns can be analyzed in this way. Some manufacturers employ polygonal rifling instead of traditional rifling, making the rifling more smooth (and thus impossible to analyze). Guns can be distinguished based on a variety of characteristics: the **gauge** of the weapon (the size of the inner diameter of the barrel, typically used with reference to shotguns), the **caliber** (diameter) of the bullet that was fired, how many **grooves** are present inside the gun barrel, and whether these grooves run **clockwise or counter-clockwise**.

Bullet Spatter Angles

More acute angle = more elongated spatter
To find impact spot, measure several angles and find point of convergence where they originated
The "tail" is actually in front, as it points in the direction of travel.

$\theta = \arcsin \frac{W}{L}$

Blood Spatter Velocity

Several mm in diameter. These may be formed by dripping from a self-inflicted or accidental wound.

Linear sequence of droplets. Typically result from blunt force injuries or when a blunt instrument covered in blood is swung (aka cast-off)

Formed from injuries such as gunshot wounds (or blunt instruments if hit with enough force). Typically less than 1 mm in diameter

Name	Symbol	Uses	Water	pH	Conductivity	Hot Water	HCl	NaOH	I ₂	Benedict's	Flame Test	Observations
Sodium Acetate	$\text{NaC}_2\text{H}_3\text{O}_2$	deicer, heating packs	VS Exothermic	8	>2000 μS	VS	NR	Does not dissolve	Yellow	NR	Yellow	Very fine white powder
Sodium Chloride Table Salt	NaCl	food, ice melt, water softener	S Exothermic	6	>2000 μS	S	NR	Dissolves slowly	Yellow	NR	Yellow	White regular crystal
Sodium Bicarbonate Baking Soda	NaHCO_3	Food	S Exothermic	8	>2000 μS	VS	Fizz	Dissolves	Yellow	Reacts to form bubbles in heat	Yellow	Very fine white powder
Sodium Carbonate	Na_2CO_3	making glass, soap, paper	Slowly VS Exothermic	10	>2000 μS	S	Fizz	Does not dissolve	Clear	NR	Yellow	White powder
Lithium Chloride	LiCl	for bipolar disorder	VS Exothermic	6	>2000 μS	VS	NR	Dissolves slowly	Yellow	NR	Red	Large Crystals
Potassium Chloride	KCl	salt substitute, CO2 laser production	VS Exothermic	6	>2000 μS	VS	NR	Dissolves slowly	Yellow	Turns green with heat	Purplish	Pinkish crystal
Calcium Sulfate Gypsum	CaSO_4	manufacture cement, sulfuric acid	VS Exothermic	6	>2000 μS	NS	NR	Does not dissolve	Yellow	NR	Does not burn	Lumpy powder
Calcium Nitrate	$\text{Ca}(\text{NO}_3)_2$	fertilizer, jet fuel, glass manufacture, metal preservative	VS Exothermic	7	>2000 μS	VS	NR	Does not dissolve	Yellow	NR	Red-orange sparks like fireworks	Huge clear crystals
Calcium Carbonate Marble Limestone	CaCO_3	diet supplement, building industry	NS	6	123 μS	NS	Fizz	Does not dissolve	Yellow	NR	Glowing bright, but does not burn	Lumpy powder, sticks to everything

Name	Symbol	Uses	Water	pH	Conductivity	Hot Water	HCl	NaOH	I ₂	Benedict's	Flame Test	Observations
Cornstarch	$\text{C}_{27}\text{H}_{48}\text{O}_{22}$	Food	NS	6	42 μS	S	NR	Clumps	Blue Black	Red brown on heating	Burns yellow	Lumpy powder
Glucose	$\text{C}_6\text{H}_{12}\text{O}_6$	Food	S Clumps	6	17 μS	VS	NR	Dissolves	Yellow	NR	Melts, burns with yellow flame	Irregular white crystal
Sucrose	$\text{C}_{12}\text{H}_{22}\text{O}_{11}$	Food	VS Exothermic	5	83 μS	VS	NR	Dissolves	Yellow	Red brown on heating	Melts, burns with yellow flame	Irregular yellowish crystals
Magnesium Sulfate Epsom salts	MgSO_4	Treating sprains	VS Exothermic	6	>2000 μS	VS	NR	Does not dissolve	Yellow	NR	Boils off with hydration, but does not burn	Cylindrical crystals
Boric Acid	H_3BO_3	Eye wash	Floats S	4	50 μS	VS	NR	Dissolves	Yellow	Bubbles on heating	Green flame	Can be either powder or crystal
Ammonium Chloride	NH_4Cl	making dry cell batteries, medical treatments	VS Exothermic	6	>2000 μS	VS	NR	Smell ammonia	Yellow	Turns dark blue immediately	Polymerizes	Irregular clearish crystal

Blood and Blood-typing

Blood can be identified at crime scenes using indicators such as luminol or phenolphthalein. These tests are not specific to blood, however, and a confirmatory test specific to blood will need to be performed if it is suspected to be present. Some exams may ask about the differences between human, avian, mammalian, and reptilian/amphibian blood. Human and mammalian blood are impossible to distinguish under a microscope, and even more sophisticated tests often give false positives. New techniques have been developed which use infrared spectroscopy to distinguish between human and mammalian blood, but they are too new to be used in the field. Human/mammal red blood cells are small and round, lacking a nucleus. Avian/reptilian blood is easy to distinguish from human/mammalian blood, as birds and other vertebrates have oval-shaped blood cells with nuclei. However, it can also be difficult to distinguish between avian and reptilian blood. Reptilian and amphibian red blood cells tend to have proportionally smaller nuclei compared to avian red blood cells, but this is not always the case. **Human Blood** has RBCs that **lack a nucleus**. Very similar to mammalian blood. **Avian Blood** has RBCs with an oval shape and **larger platelets** than human blood. **Mammalian Blood** has RBCs that also lack a nucleus and appears identical to human blood. **Reptilian/Amphibian Blood** has RBCs with an oval shape and may have irregular borders or irregular nuclei.

Blood typing is a system of categorizing blood based on proteins called **antigens** which are present on the surface of red blood cells. There are 44 different human blood group systems recognized internationally, but the two most important ones are ABO and Rh. The ABO blood group system deals with the presence or absence of **A and B antigens**, while the Rh blood group system mostly deals with the presence or absence of a **Rh(D)** antigen. There are **four blood types** seen in humans using the ABO system: **A, B, AB, and O**. These are named for the antigens they possess: A has A antigen, B has B antigen, AB has both A and B antigens, and O has neither antigen. Each of these blood types can also be positive or negative for the Rh(D) antigen (also known as the rhesus or Rh factor). Combining the two systems means that there are **eight total blood types: A+, A-, B+, B-, AB+, AB-, O+, and O-**. The rarest blood type is AB-, while the most common blood type is O+. Since antigens are what determine someone's blood type, you can determine the type of blood that someone has using **antibodies**. Antibodies are proteins made by the immune system designed to keep foreign entities out of the body. There are antibodies which will react to certain antigens present on the surface of the red blood cell, causing blood to **agglutinate**. Agglutination might occur naturally if someone was given the wrong blood type in a blood transfusion. The antibody detects the antigen, causing the blood cells to clump together. In this case, seeing clumps of blood will confirm that the antigen you're testing for is present on the blood cell. Examine the chart "How to Read Your Results". Since Anti-A antibodies are designed to interact with the A antigen, type A and AB blood will agglutinate when exposed to Anti-A serum. Similarly, since Anti-B antibodies will interact with the B antigen, both B and AB will agglutinate when exposed to Anti-B serum. Since type O blood lacks both the A and B antigens, it will not agglutinate when exposed to either serum. If a lab scenario uses a simulated blood kit, there may be plates or similar pieces of equipment to perform the tests. If not, use a well plate. Place a small amount of the blood sample in a well, and then place a small amount of the serum. Wait a moment for the "blood" to react, and then record the results. Be sure to keep track of which serums are placed in each well. Some simulated blood typing kits will only have Anti-A and Anti-B serums, but some will also include an Anti-Rh or Anti-D serum. This behaves similarly to the Anti-A and Anti-B serums in that it will cause cells that have the Rh(D) antigen to agglutinate. If a blood sample clumps up when exposed to this serum, the blood type is positive. If it does not, the blood type is negative.

Hair is typically divided into two main parts: the hair shaft (the fiber that emerges from the skin) and the hair follicle (which is embedded in the skin). The hair follicle is also known as the bulb or root when it is removed from the skin, and is responsible for the growth of hair. The hair follicle is the only part of hair considered to be alive, since it is the site of all the biochemical activity in the hair. There are also other structures associated with the follicle such as sebaceous (oil-producing) glands and muscles which make the hair stand on end. The shaft can be further divided into three layers called the cuticle, cortex, and medulla. The **medulla** is the innermost layer of the hair, and may or may not be present in some hairs. This region of the hair lacks the same structure that is present in the outer layers, and it is one of the most fragile layers of hair. The role the medulla plays in the hair is unclear, but recognizing certain characteristics of the medulla is essential to identifying hairs. Human hairs can have three different medulla types: fragmented, interrupted, or continuous. Fragmented medullas are the most broken pattern, and appear more like a dashed line with many gaps and fragments. Interrupted medullas will have breaks, but will be much less broken than fragmented medullas. A majority of the medulla is connected, and the gaps are small. Continuous medullas have no breaks. Animal hairs typically have thicker medullas, and can have two additional patterns called ladder or lattice. Hairs can also be characterized by the medullary index. This is measured by taking the diameter of the medulla and dividing it by the diameter of the hair as a whole. Humans typically have a medullary index of less than 0.3, while animals typically have a medullary index of greater than 0.5. The **cortex** is the second layer of hair, and it is the most structurally complex of the three. The cortex gives hair its color and shape, and is also responsible for water uptake and nourishing the hair. The pigment that gives hair its color is melanin, which is the same pigment responsible for coloring skin. Melanin is found in pigment granules, which in humans tend to be distributed towards the cortex. In animals, most of the pigment is found closer to the medulla. The cortex also contains oil-filled pockets, which are oval shaped structures commonly found in cattle and dog hairs (though they may also be found in human hairs). Cortical fusi are irregular oval pockets found near the root of human hair, though they may be present elsewhere in the shaft. The **cuticle** is the outermost layer of the hair, and is responsible for the scale-like pattern on the outside of a strand of hair. The cuticle of the hair is responsible for protecting the inner layers and repelling water. The cuticle can have a variety of patterns that are useful for identifying hairs. Coronal scales are present on the hair of bats and rats, and they look like stacked cups or "strawberries" on a stick*. Spinous scales are present on cat hairs, and have points that are rounded at the ends. Imbricate scales are found on human hairs, where the scales are more rectangular and flat in shape. Many mammals also have hairs with imbricate cuticles. **Hair grows in three stages: the anagen, catagen, and telogen phase**. Occasionally a fourth phase is included known as the **exogen phase**, but this is largely just an extension of the telogen phase. The **anagen phase** is the first phase, and is also known as the **growth phase**. In this phase the hair grows around 1 cm per month for around three to five years. A majority of hairs (around 85-90%) on the head are in the anagen phase. The next phase is the **catagen phase**, also known as the **transitional phase**. This phase lasts around two weeks, during which the follicle shrinks and the hair is cut off from its blood supply. This forms a club hair, and causes the hair to enter the **telogen phase**. In this phase (also known as the **resting phase**), the hair is dormant and anchored in by epidermal cells lining the follicle. The follicle will eventually begin to grow again, causing the anchor point of the shaft to soften and the hair to be shed. The exogen phase is the process of shedding the hair, while the telogen phase is the hair laying dormant.

PCR

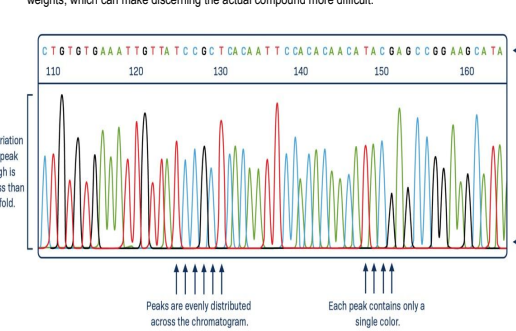
DNA template with sequence of interest
Primers
Taq Polymerase

1st cycle
2nd cycle
nth cycle

PCR product
~2ⁿ copies

- **Human Hair** has a scaly, imbricate cuticle and its medulla is typically fragmented or interrupted; may lack medulla entirely
- **Squirrel hair** has a thick, latticed medulla
- **Cow Hair** is very coarse, thick. Ovoid bodies are present in the cortex. Thinner medulla than horse hair, and may be fragmented or interrupted.
- **Horse Hair** has an imbricate cuticle. It is very coarse and thick
- **Bat Hair** is very fine and has coronal scales on the cuticle, similar to a stack of paper cups; "Strawberries on a stick"

Reading a Mass Spectrum: A mass spectrum is the output of a mass spectrometer. It is a graph with two axes: the x-axis represents the **mass to charge ratio** (m/z), and the y-axis represents the **relative abundance** of each detected ion. Each line on the graph is known as a **peak**. The graph is known as the **base peak**. This is often also the **molecular ion peak**, but not always. The molecular ion peak is simply the peak represented by the molecular ion, which is the ion formed by removing one electron from the original compound. Since the mass of the electron is so small, the mass of the molecular ion is nearly identical to the original compound. This peak is also known as the (M)⁺ peak, and provides the weight of the original compound. This is the mass spectrum of methane, CH₄ (up). The tallest peak is located at 16. This means that the base peak is located at 16. It has a relative abundance of 100%, meaning that the height of every other peak is described relative to the base peak. The base peak in this case is also the molecular ion peak, though this is often not the case for large molecules. This mass spectrum also has peaks located at 15, 14, 13, and 12. Since methane is a small molecule, there are only a few ways for it to fragment and produce ions. Losing one hydrogen atom produces a fragment with a $m/z = 15$, which can then lose another hydrogen atom to produce a peak at $m/z = 14$. This continues until all 4 hydrogen atoms have been lost, leaving peaks from 12 to 15. **Isotope Peaks:** It might seem like every peak is accounted for. However, there is still a small peak at $m/z = 17$. The molecular ion peak, or (M)⁺ peak is at 16, since 16 is the mass of methane. How can a peak be present with a m/z greater than that of the molecular ion? The peak at 17 is due to the presence of carbon-13, an isotope of carbon. Isotopes are different forms of elements that have the same number of protons but different numbers of neutrons. 98.93% of carbon atoms are carbon-12, the isotope of carbon with 6 protons and 6 neutrons. Carbon-13 (also written as ¹³C) has 7 neutrons, occurring with an abundance of 1.1%. This means that while 98.93% of the ions contain carbon-12, 1.1% of them will contain carbon-13. Methane molecules which incorporate carbon-13 will have a mass of 17, which is why there is a small peak at $m/z = 17$. This peak is known as the ($M+1$)⁺ peak, since the m/z is one greater than the (M)⁺ peak. **Degrees of Unsaturation** (also known as the hydrogen deficiency index) of an atom indicates how many double bonds or aromatic rings are present in its structure. Without this information, it is nearly impossible to predict the structure of a compound from its formula. Many chemicals will share a common molecular formula, but the degrees of unsaturation can help distinguish between them. The formula for degrees of unsaturation is $DoU = (2C + 2 - H - X) / 2$ where C is the number of carbons, H is the number of hydrogens, and X is the number of halogens. For example, for the compound C₁₁H₈Cl₂O the degrees of unsaturation is 7. This means that there are 7 double bonds or ring structures in the compound. A molecule with a DoU of 0 cannot have any double bonds or rings, and a molecule with a DoU of 1 can have a double bond or a ring, but not both. **Example:** The first step to identifying the compound depicted is **locating the molecular ion peak**. This is most likely the tallest fragment with the highest m/z . For this compound, the molecular ion peak is located at $m/z = 86$. This means the molecular weight of the compound is 86. The molecular weight can say a lot about a compound, such as whether or not it contains nitrogen. Compounds with an even molecular weight likely do not contain nitrogen or have an even number of nitrogen atoms. Compounds with an odd molecular weight likely contain an odd number of nitrogen atoms. Since the molecular weight of this compound is even, it is too soon to say whether or not it contains nitrogen. **Calculate the number of carbons in the compound.** This is done by comparing the (M)⁺ and ($M+1$)⁺ peaks. It's difficult to tell with this scale, but the relative abundance of the ($M+1$)⁺ peak is around 1.2%. However, the molecular ion peak is not the base peak. The base peak appears at $m/z = 43$. While the ($M+1$)⁺ peak is around 1.2% as tall as the base peak, you want to compare the ($M+1$)⁺ peak to the (M)⁺ peak. To do this, divide by the relative abundance of the molecular ion peak before multiplying by 100%. Since the relative abundance of the (M)⁺ peak is around 21, the ($M+1$)⁺ peak is around 5.7% as tall as the molecular ion peak. Dividing this number by 1.1 for the abundance of carbon-13 gives a value of 5.2. Rounding to the nearest whole number indicates that there are 5 carbon atoms in this molecule. **Examine the ($M+2$)⁺ peak.** This compound doesn't have a ($M+2$)⁺ peak, so it likely does not contain a halogen like chlorine or bromine. **Analyze the mass of the molecule.** By inspecting the molecular ion peak, we know that this unknown molecule weighs 86 amu. In step 2, we determined that there are likely 5 carbon atoms present in this molecule. These 5 carbons weigh 12 amu each, meaning that they weigh 60 amu in total. Subtracting 60 from 86 means that the rest of the atoms in the molecule weigh 26 amu total. There's no way to draw a structure which has 5 carbon atoms and 26 hydrogen atoms, so that means there must be another element present. **Identify any heteroatoms.** Heteroatoms are elements other than carbon or hydrogen. Remember, we ruled out any ($M+2$)⁺ atoms in step 3. The most common heteroatoms are nitrogen and oxygen. However, in step 1 we established that the compound likely does not contain nitrogen as it has an even molecular weight. This means that the compound likely has one oxygen atom. **Put it all together.** The molecular weight of the original compound as defined by the molecular ion peak is 86 amu. In step 2 we calculated that there are likely 5 carbons in the compound weighing 60 amu total. Determining that there is one oxygen atom in the compound which weighs 16 amu means that there are 10 hydrogen atoms in the compound. As a result, this compound's molecular formula is most likely C₅H₁₀O. Remember, it is often **not possible** to determine the exact molecular formula of a compound solely given its mass spectrum. It depends on the amount of information given and the possible fragments the compound can produce. Many potential fragments can have the same or very similar molecular weights, which can make discerning the actual compound more difficult.

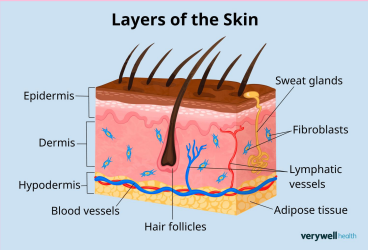


On a chromatogram, each peak represents the detection of a distinct chemical compound that has been separated from a mixture during the chromatographic process, with the height or area of the peak indicating the relative amount of that compound present in the sample;

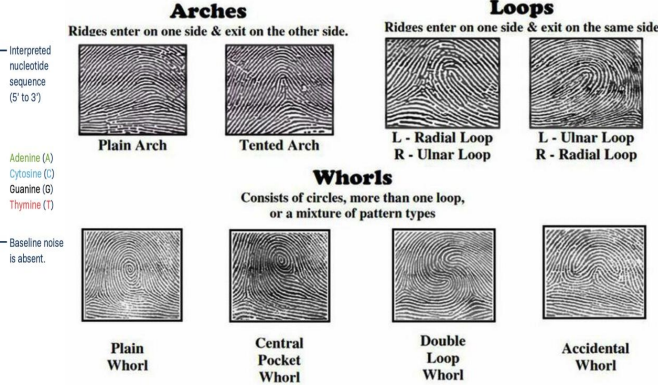
Fingerprint identification: ACE-V

- Analysis:** examiner determine if fingerprint is suitable for use in a comparison.
- Comparison:** a known print will be compared with the suspect print. Known prints may be collected from suspects, or pulled directly from databases like IAFIS. IAFIS (Integrated Automated Fingerprint Identification System) is a database used by the FBI starting in 1999 to store and organize fingerprints collected in the United States. Since 2011, the FBI has made use of NGI (Next Generation Identification) instead of IAFIS.
- Evaluation:** the examiner will evaluate if they're from same source
- Verification:** another examiner independently analyzes the prints and either supports or refutes the claims of the original examiner.

Skin: the epidermis, the dermis, and the hypodermis. The epidermis is the outermost layer of skin, while the hypodermis is the deepest layer of skin. There are two types of skin, thick and thin skin. **Thick skin** is found on the fingertips and palms of the hands and soles of the feet. This is the skin that fingerprints are made of. The most important layer of the skin in the context of fingerprints is the epidermis, which is the outer layer of skin. Cells begin their life in the basal layer and are constantly pushed upwards, moving through the layers of skin. This ensures that dead skin cells can be shed and replaced. The epidermis can be further divided into five layers in thick skin: Stratum corneum (cornified layer) is the outermost layer of the epidermis. It is formed from dead cells almost entirely filled with keratin. These cells are continuously shed and replaced. Stratum lucidum (clear layer) is an additional layer only present in thick skin. Stratum granulosum (granular layer) is where keratin filaments that were formed in the stratum spinosum are bound together. Cells begin to die once they reach this layer of the skin, as they no longer receive nutrients from the capillaries in deeper layers. Stratum spinosum (spiny layer) is a layer featuring cells bound together in structures called desmosomes. Keratin production begins when cells are in this layer. Stratum basale (basal layer) is the innermost layer of the epidermis. This is the layer responsible for fingerprint formation in the womb. When the basal cells present in this layer form ridges, small projections called epidermal ridges form. This also helps the epidermis obtain nutrients from the dermis.



Forensic Entomology is the study of insects found on a dead body to help long since death. Most common = beetle and blowfly. Blowflies appear first, within minutes or hours of the death. Flesh flies can arrive at the same time as blowflies but generally arrive later. Certain amounts of time lapse between each life stage, which can tell this time. For example, if only maggots were found on the dead body, that means the victim probably died less than 24 hrs ago. Beetles usually arrive well after the blow and flesh flies and are generally the last insect left on the body after months of decomposition. Mites are also generally present with these beetles initially because they help suppress maggots, and as such allow certain types of beetles.



Development

- Dusting:** Powder applied to prints sticks to fatty acids and lipids. Generally, this method involves using a special brush, usually made of camel hair, to lightly spread the powder over the area where prints may be found, usually smooth or nonporous surfaces. There are numerous different fingerprinting powders used in dusting, and their usages vary depending on the surface and the scene environment. For example, it would make more sense to use a dark-colored powder on a light-colored surface or a fluorescent powder on a dark-colored surface. The exact compositions of such powders vary, as most formulas are kept proprietary by their manufacturers
- Iodine Fuming:** One of the earliest methods of fingerprint development, this method uses iodine to visualize latent fingerprints. It is fast and inexpensive, reacting with body fats and oils in prints. Iodine is a solid but it easily sublimates, forming a dark purple vapor. By placing the fingerprint in a chamber with some iodine, the iodine vapor will adhere to the oils in the fingerprint. This produces a temporary fingerprint that will fade in a few days. To "fix" the fingerprint and keep it permanently, the fingerprint is then treated with a starch solution which reacts with the iodine and turns black. This method of development works well on surfaces like paper or cardboard.
- Ninhydrin:** A chemical method that is useful for lifting latent prints on paper. The object with the print is dipped in or sprayed with ninhydrin, then left to develop for 24 hours. It reacts with amino acids left behind in prints and results in purple fingerprints.
- Cyanoacrylate (Superglue) Fuming:** This form of development utilizes cyanoacrylate (CA), also known as superglue. A special chamber is used, and the CA is heated in order to form a vapor. This vapor will react with moisture and organic molecules in the fingerprint, forming a white fingerprint. This works best on nonporous surfaces like glass, plastic, and metal.
- Other Methods: Silver Nitrate:** The object with the fingerprint is dipped in or sprayed with silver nitrate, which reacts with chlorides in the fingerprint and turns grey when exposed to light. This technique is commonly used for surfaces like wood or styrofoam. **Small Particle Reagent (SPR):** Not as common as the other methods used, but still important. SPR is used for wet surfaces and reacts with the lipids present in fingerprints. **Wetpup:** a special pre-mixed liquid formula that is designed to lift latent prints on the sticky sides of adhesive surfaces (i.e. most kinds of tape) **Alternate Light Sources:** Lasers and LEDs in combination with certain filters or dyes are used to reveal the location and pattern of fingerprints. **Columnar Thin Film (CTF):** Deposits a thin-film coating on the fingerprint in a specific chamber, allowing for preservation of the ridges and topography of the fingerprint.

Fractures: Another way pieces of glass evidence can be evaluated is by examining any fractures that are present. Glass forms two main types of fracture lines: radial lines and concentric lines. Radial lines are the first to form, and occur first on the opposite side to the impact (e.g. if the impact came from the inside, the outside would have failed). These fractures "radiate" out from the site of the impact like spokes on a bike wheel. Concentric fracture lines appear first on the same side as the impact, forming a circle around the breaking point. Cracks will always end at existing cracks. This information can be used to determine the order in which fractures occurred. In the image to the left two fractures are present: fracture A and fracture B. The radial fracture lines from fracture B are stopped by fracture lines from fracture A. This means that fracture A was present before fracture B was formed. Stress lines on the edge of a piece of glass will also provide information as to what direction the force came from. Remember the 3R rule: radial fractures make right angles to the reverse side of impact.

Reflection and Refraction:

- angle of incidence = angle of reflection
- index of refraction → $n = c/v$ or speed of light in vacuum/speed in medium
- Snell's law to find refractive index (see orange box)

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

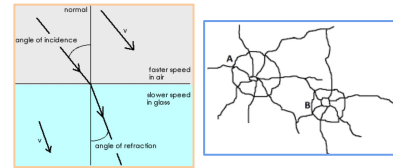
Pollen and seeds (produced by spermatophytes)

- The anther (on top of the stamen) produces pollen
- Then it is transferred to the stigma of the same flower (self-pol) or a different one (cross-pol)

- Pollen has 3 parts: an inner cytoplasmic portion which contains the nuclei, an inner wall called an intine or endospore, and an outer wall called an exine or exospore. The intine is made up of cellulose, similar to the cell wall in plants. The exine, however, is made up of an incredibly tough compound called sporopollenin, which is resistant to degradation.

- Since it is so hardy, this makes pollen evidence very difficult to wash off or get rid of. Pollen grains are very good at attaching to surfaces

- This is also due in part to the unique structure of the exine, which is visible in microscope images. Exine structures differ wildly between different plant species, and by studying the structure of the pollen grain it is possible to determine what type of plant it came from. Forensic use of pollen is forensic palynology.



Glass type	Density (g/cm³)
Fused silica	2.18
Borosilicate glass	2.2-2.5
Soda-lime glass	2.4-2.8
Lead crystal	3.1

